

**The profitability of simple technical trading strategies:
the case of Hong Kong**

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ABSTRACT

In this paper, we investigate two simple but very popular trading rules, simple moving average (SMA) and trading range break (TRB), by looking at a series of data of Hang Seng Index over thirty-five years, from 1972 to 2006. To avoid the data-snooping biases, we apply the out-of-sample tests in addition to the traditional tests to check for the validity of the profitability of the trading rules.

In general, our results show that there is one trading rule, the (1, 50) rule, outperforms the market (HSI) over the 35 years testing period and in the pre-1986 sub-periods in our study in the SMA trading rule. These average returns from SMA rule are higher than the returns found in the US market (Brock et al., 1992) and Hong Kong market (Bessembinder and Chan, 1995). Besides, the VMA (1, 50) rule performs better than the FMA (1, 50) rule because it includes the information of the first nine days into investors' decision. With the flexibility of buy or sell within the first nine days, the VMA (1, 50) rule can generate 2.5% to 7.5% (annual) more profit than the FMA rule before transaction costs. However, the returns of the TRB rules are all small and insignificant.

We find that the high trading returns are unlikely to be explained by the difference risk in difference trading strategies and transaction costs. However, they can be partly explained by the positive first order autocorrelation in the Hong Kong stock market and the time-varying premium. Furthermore, the inconsistent subperiod abnormal average returns suggest that the stock market integration leads to a more efficient dissemination of information over time in Hong Kong.

Keywords: Technical trading rules; Variable Moving Average; Fixed Moving Average; Trading Range Break

1. INTRODUCTION

As one of the most volatile stock markets in the world, the Hong Kong stock market provides ample of profitable opportunities to investors. Can investors make profit using technical trading strategies in the Hong Kong stock market? The answer to this question is an important but still unanswered question to investors in the Hong Kong stock market, the Hong Kong Exchange and Clearing Limited (HKEX). The HKEX is one of the leading and important stock exchanges in the world and Asia. It ranked sixth in the world and second in Asia in terms of market capitalization at the end of 2006.

There has been some research done on the profitability of using trading strategies in the HKEX (e.g., Bessembinder and Chan (1995) and Coutts and Cheung (2000)). These studies have two distinct features. First they use linear models and second they find no significant profits after transaction costs. However, there is no published studies in exploiting the use of linear model using the out-of-sample test, a method free from selection bias, to find profitable trading strategies so far. The main purpose of this paper is to utilize the out-of-sample test to investigate whether there is any chance of making abnormal profit using existing technical trading strategies or trading rules in the Hong Kong stock market.

The main reason that we use the out-of-sample test in this study is to avoid the well-known data-snooping problem. According to Lo and MacKinlay (1990), the more scrutiny a data set is received, the more likely that interesting spurious patterns will be uncovered¹. To avoid the data-snooping biases problem, our study examines the trading rules using both bootstrap and out-of-sample tests. By using the out-of-sample test, which applies trading rules in one period to another period, we can compute test statistics from a different data period. Thus, the interesting patterns found will be less likely affected by the data-snooping biases. The testing results of the out-of-sample test would then provide valuable insight to the results of the traditional and bootstrap tests.

¹ According to Lo and MacKinlay (1990), the data-snooping biases occur if test statistics are affected by the empirical relations uncovered in the same data that test statistics are applied. The new trading rules may exhibit apparently significant forecast power simply because the same data are used to test the new trading rules. In such a case, researchers may mistakenly conclude significant trading rule profits because of the data-snooping biases.

In this paper, we investigate two simple but very popular trading rules, simple moving average (SMA) and trading range break (TRB). We study the trading rules by looking at a series of data of Hang Seng Index over thirty-five years, from 1972 to 2006. Our results show that there is one trading rule, the (1, 50) rule, consistently outperforms the market (HSI) over the 35 years testing period and the four sub-periods in our study in the SMA trading rule. These average returns from SMA rule are much higher than the returns found in the US market (Brock et al., 1992) and Hong Kong market (Bessembinder and Chan, 1995). For the TRB rules, the 50 days TRB rule can also provide us with consistent trading profits. Besides, the variable moving average (VMA) (1, 50, 0) rule outperforms all the other rules by at least three times. From a practical point of view, the VMA (1, 50, 0) rule should be the best trading rule, among the rules we study in this paper, to be used by investors in the Hong Kong stock market.

The observation that trading strategies generate high abnormal average returns seems to suggest that the Hong Kong stock market is inefficient. This conclusion is a bit premature and, therefore, we further explore four alternative explanations for the observation. First, using out-of-sample test, we show that it is unlikely that the difference risk in difference trading strategies can explain the abnormal returns. Second, we find that the abnormal high average returns from the SMA and TRB rules can be partly explained by the positive first order autocorrelation in the Hong Kong stock market. Third, we find that the average returns are still very high after transaction costs. So, transaction costs do not explain the high average returns either. Furthermore, the high abnormal average returns can be due to the consequence of stock market integration which leads to a more efficient dissemination of information. This study contributes to the literature by testing a more recent data and using improved methodology to test the Hong Kong stock market.

The paper is organized as follows. The next section will discuss literature review of market efficient hypothesis. Section 3 will present the data and methodology. Empirical results will be reported in section 4. The last section, section 5, will summarize and conclude this paper.

2. LITERATURE REVIEW

In his seminal paper, Fama (1970) classifies the Efficient Market Hypothesis (EMH) into three forms, weak-form, semi-strong form and strong form, according to the types of information to be reflected. The weak-form market efficiency hypothesis states that today's security prices fully reflect all information contained in historical security prices. That means we cannot find any pattern in security prices or security prices have no "memory". The hypothesis is concerned with the forecasting power of past returns. Therefore, the use of today's stock price as a forecast of tomorrow's stock is as likely to be as good as any forecasting tool. This implies that, if the market is weak-form efficient, no investor can earn excess risk-adjusted returns by developing trading rules based on historical price or return information.

The benefit of trading strategies has been studied in both developed and developing countries. In US, early studies show that filter rules can earn small abnormal returns but they vanish after taking into account transaction costs (Alexander (1961, 1964)), Fama and Blume (1966) and Sweeney (1988)). Neftci (1991) evaluates *ad hoc* trading rules on the Dow Jones Index and finds that few of the rules generate well-defined techniques of forecasting. However, Brock Lakonishok and LeBaron (1992) use a long period of data, over 100 years, to test on two technical trading rules (moving average and trading range break) via bootstrapping. They find that these trading rules contain predictive ability. However, Bessembinder and Chan (1998) document that the inclusion of trading costs and adjustments for non-synchronous trading eliminates the technical trading rule profits in U.S., but not the economic significance of the Brock et al. study. Blume, Easley and O'Hara (1994) are able to establish profitable trading rules with trading volumes. They find that traders who use information contained in market statistics do better than traders who do not.

In U.K., Hudson, Dempsey and Keasey (1996) replicate Brock et al.'s (1992) research and find that, in the U.K. market, the trading rules have predictive ability if sufficiently long series of the stock indexes are considered. However, excess returns vanish with the presence of trading costs. Mills (1997) also reaches the same results as Hudson *et al.* (1996). Isakov and Hollistein (1998) find that transaction costs eliminate technical trading profits in the Swiss stock market.

In Asia, Bessembinder and Chan (1995) replicate Brock et al.'s (1992) study in six Asian markets. They find that the trading rules are quite successful in the emerging markets of Malaysia, Thailand and Taiwan but have less explanatory power in more developed markets such as Hong Kong and Japan. Ratner and Leal (1999) examines the predictability of the Variable Length Moving Average (VMA) trading rule in ten emerging Latin American and Asia stock markets (from 1982 to 1995) and find that the VMA trading rule may be profitable in Taiwan, Thailand and Mexico. Coutts and Cheung (2000) study the trading rule profitability in the Hong Kong stock market and find that the rules fail to provide positive abnormal returns after transaction costs.

The profitability of trading strategies may be associated with irregularities in the markets. For example, in US studies find that there is positive autocorrelation of weekly returns on portfolio of NYSE stocks (Lo and MacKinlay (1988) and Conrad and Kaul (1989)) and negative autocorrelation on individual stocks (Fama and French (1988), Poterba and Summers (1988), and Lehmann (1990)). Jegadeesh (1990) finds negative serial correlation for lags up to two months and positive correlation for longer lags. Other market inefficiencies are also reported. They are namely, the market overreaction (DeBondt and Thaler (1985, 1987) and Chopra, Lakonishok, and Ritter (1992)), the January effect (Rozeff and Kinney (1976) and Ritter and Chopra (1989)), the Monthly effect (Ariel (1987) and Lakonishok and Smidt (1988)), the Weekend effect (French (1980) and Keim and Stambaugh (1984)) and the holiday effect (Lakonishok and Smidt (1988) and Ariel (1990)). These indicate that stock prices do not necessary follow random walk in US. The findings of the literature seem to show that the daily and weekly stock returns can be predicted from past stock returns and financial variables.

In Europe, evidences on market inefficiency are also found. For example, autocorrelation is found in the Swedish stock market (Frennberg and Hansson (1993)), seasonality and weekend effect are found in the London Stock Exchange (Reinganum and Shapiro (1987) and Jaffe and Westerfield (1985)), and weak-form inefficiency is found in the Athens Stock Exchange (Spyrou (1998), Kourtmos, Negakis and Theodossiou (1993), and Kavussanos and Dockery (2001)).

In Asian stock markets studies show that non-trivial first order autocorrections are found in six stock markets, namely, Hong Kong, Japan, Korea, Malaysia, Thailand and

Taiwan (Bailey, Stulz and Yen (1990), Pan, Chiou, Hocking and Rim (1991), and Bessembinder and Chan (1995)). Besides, the January Effect (Gultekin and Gultekin (1983) and Kato and Schallheim (1985)), the Weekend Effect (Lee, Pettit and Swankoski (1990), Ho (1990), Wong, Hui and Chan (1992) and Ho and Cheung (1994)), and the Chinese New Year Effect (McGuinness (2005)) are found in various Asian countries (for example, Japan, Korea, Malaysia, Thailand and Taiwan).

In sum, the evidence on runs tests and random walk tests suggests that US market is weak-form efficient over a short horizon, but ambiguous over longer horizons. However, findings related to trading rules based on technical analysis suggest that stock returns are predictable from past returns. This evidence is contrary to that found in random walk tests and seems not to support the weak-form efficiency hypothesis in the US market. In Europe, the evidence suggests that excess returns can be generated by trading rules but they vanish after transaction costs. In Asian markets, however, test results indicate that most of the major markets are weak-form inefficient in both individual stocks and portfolios of stock. In Hong Kong, studies find positive abnormal returns using the two simple trading rules but they vanished after transaction costs.

3. DATA AND METHODOLOGY

A) Data

We are using daily closing value of Hang Seng Index (HSI) in our study. The data period is from 1st January 1972 to 31st December 2006, totally 35 years. The data is abstracted from the DataStream database available in the University of Macau.

In order to investigate the consistency of the trading strategies, we sub-divide the data into four sub-periods. The first sub-period is from 1st January 1972 to 31st December 1978 (7 years). The second sub-period is from 1st January 1979 to 31st December 1985 (7 years). The third sub-period is from 1st January 1986 to 31st December 1995 (10 years) and the fourth and last sub-period is from 1st January 1996 to 31st December 2006 (11 years).

To sub-divide the data set, we use 1986, the year where four Hong Kong stock exchanges unified into one stock exchange, the Stock Exchange of Hong Kong, as the cutoff year. We then further sub-divide the two periods into two equal sub-periods.

One problem with the HSI is that it does not adjust for dividend payment. Therefore, it does not contain information about the payment of dividends and may cause bias in the study. However, Mills and Coutts (1995), after reviewing literatures concerning dividends, conclude that the bias in excluding dividends is minimal, a conclusion confirmed by Draper and Paudyal (1997).

The HSI is a value weighted index composed of 33 actively traded “blue chip” stocks. Since the constitute stocks in HSI is very actively traded stocks, there is no “thin trading” and “non-synchronous” trading problem in the index (Ho and Cheung, 1991). Besides, the HSI reflects a broad industrial base and represents more than 70% of the market value of the HKEX. Therefore, it is an ideal index for investigating the validity and applicability of trading rules.

B) Methodology

In technical analysis, there are many techniques designed to identify and exploit patterns in historical prices. In this paper, we focus on simple moving average (SMA) and trading range break (TRB) rules because the buy and sell signals generated from the two trading rules are unambiguous. Other technical trading rules, such as “head and shoulders” patterns and Elliot Waves, cannot be tested properly because they require trader interpretation which lead to unambiguous buy and sell signals. Another reason for using the SMA and TRB rules is because we want to compare our results with previous studies. So, in our study, we examine the same set of trading rules, SMA and TRB rules, used by Brock et al. (1992) and Bessembinder and Chan (1995).

B. 1) Single Moving Average (SMA) Trading Rule

The SMA can be constructed by summing the data for a specific period (e.g., 10 days) and take the average. It is called the n-day moving average. If the moving average is 20 days or below, it is regarded as a short-term moving average. If it is between 20 and 50 days, it is a medium-term moving average. If it is greater than 50 days, it is a long-term moving average. The rule is that if the short-term moving average breaks the medium or long-term moving average from below, it sends out a buying signal. It is because, when short-term moving average breaks from below, the market must be rising.

It is expected that the market will keep on rising for a while. The shorter moving average tends to have larger values than the longer average, requiring a long position. Thus, it is time to buy at the beginning of the rising market trend. A selling signal emerges when the short-term moving average breaks the medium or long-term moving average from above. It signals that the market is falling. When the market is falling, the shorter average tends to have lower values than the longer average, requiring a short position.

The SMA trading rules are further divided into variable-length moving average (VMA) rules and fixed-length moving average (FMA) rules. For the VMA rules, buy (sell) signals are generated when the short-term moving average exceeds (falls below) the long-term moving average by a prespecified percentage band. If it is inside the bandwidth, no signal will be generated. If the band is 0%, the VMA rules classify every day into either buy or sell day. The use of bandwidth is to avoid the emission of false signals when the short-term and long-term moving averages are close to each other.

The FMA rules use the same set of rule as VMA rules to generate buy (sell) signals. However, the FMA rules further assume that returns should be different for a few days after the signals are generated. Thus, if signals are generated, the FMA rules require investors to stay in the same position (i.e., either buy or sell) for a fixed number of days, 10 days in this study. Other signals generated during this 10-day period are ignored. When the 10-day period passes, the FMA rules start to react to new signals.

For each of the VMA and FMA rule groups, this study evaluates the five variations of the rules, (1, 50), (1, 150), (1, 200), (2, 200) and (5, 150), where the first number in the parentheses denotes the number of days for the short-term moving average and the second number denotes the number of days for the long-term moving average. In addition, each rule is evaluated with the bands of 0%, 1%, 2%, and 3%, making for 20 individual rules in total for each rule group.

The buy and sell signals are generated as follows:

$$\Sigma R_{i,t}/S > (1 + X)\Sigma R_{i,t-1}/L = \text{Buy}$$

and $\Sigma R_{i,t}/S < ((1 + X)\Sigma R_{i,t-1}/L = \text{Sell}$

$$\text{where } R_{i,t} = (I_{i,t} - I_{i,t-1}) / I_{i,t-1}, \text{ and } X = 0\%, 1\%, 2\% \text{ or } 3\%$$

R_{it} is the daily return of Hang Seng Index in period S (1, 2, or 5 days); $R_{i,t-1}$ is the daily return of Hang Seng Index over period L (50, 150, and 200 days); $I_{i,t}$ is the closing Hang Seng Index at t ; X is the prespecified percentage band.

In this paper, we use the out-of-sample simulations to investigate whether the trading rule works in general and also to identify which combination(s) works best in particular. The problem with the traditional approach of simulating profit from historical period and choosing one best SMA is that it is subjected to selection bias, if enough combinations are tried, at least one SMA is likely to work simply by chance. The advantage of using the out-of-sample simulations is that the results are not contaminated by selection bias. We follow a two-step procedure to select optimal SMAs and their profits. We start by searching for the most profitable SMA among the 5 possible combinations of SMAs within a particular bandwidth for Year 1 of our sample. We then use that optimal combination (e.g., 1 and 50 days moving averages or (1, 50, 0) trading rule) to simulate trading in Year 2. The simulation is done by applying the trading rule (e.g., (1, 50, 0)) in Year 1 to Year 2 HSI data and then compute and record the returns generated. The next step is to reoptimize by searching for the most profitable SMA for Years 1 and 2 combined. We then use the new optimal combination (e.g., 2 and 200 days moving averages) to simulate and record returns for Year 3. This sequential searching for the optimal SMA and simulation of return is continued through the end of the sample period.

We have 37 years of data (1970 – 2006, including the 200 observations for Year 1) and have 35 years (1972 – 2006) of out-of-sample daily returns simulations from following a moving average trading strategy. We measure returns on stock prices as the difference between closing price on Day t (P_t) and the closing price on Day $t - 1$ (P_{t-1}) divided by the closing price on Day $t - 1$ (P_{t-1}). This statistic is then multiplied by 1 if the trading rule specifies a long position and multiplied by -1 if the rule requires a short position. To determine whether the simple moving average trading rule generates abnormal returns requires a risk-adjusted measure of return to compare with alternative investments. We use the return from the buy-and-hold strategy for the HSI as our benchmark measure.

B. 2) Trading Range Break (TRB) Trading Rule

In contrast to the simple moving average rule, the second trading rule, the trading range break (TRB), initiates a sell (buy) signal if the security price falls below (rises above) some pre-defined support (resistance) level, the highest (lowest) price of the security in the previous period. The time periods are 50, 150 and 200 days and the bands are 0%, 1%, 2% and 3% respectively. The 10-day cumulative returns after the signals will be reported. The TRB rule is similar to the SMA rule by identifying the trend in price series. An upward (downward) trend is identified when the current price is equal or greater (less) than the maximum (minimum) of prices over the period of the prespecified length. A long (short) position is taken when the stock price breaks the maximum (minimum) barrier. The logic for this rule is that when the current price reaches the previous peak (low), a great deal of selling (buying) pressure arises because many people would like to sell (buy) at the peak (trough). However, if the price exceeds the previous peak (low), it is indicated that the upward (downward) trend has initiated. The formulas for the buy and sell signals are as follows:

$$P_t > (1 + X)\max(P_{t-1}, \dots, P_{t-m}) = \text{Buy} \quad \text{and}$$

$$P_t < (1 - X)\min(P_{t-1}, \dots, P_{t-m}) = \text{Sell}$$

where P_t is the closing HSI closing at t ; m is the time period we selected (50, 100 and 200 days); X is the percentage band, 0%, 1%, 2% and 3% respectively; $\max(P_{t-1}, \dots, P_{t-m})$ is the maximization operator for the Hang Seng Index series.

This study evaluates the TRB rules where recent maximums and minimums are defined as the extreme observations over the time period 50, 150 and 200 days, respectively. In addition, each rule is evaluated with the bands of 0%, 1%, 2%, and 3%, making for 15 rules in total. Similar to the SMA trading rules, the benchmark for evaluating the abnormal returns for the TRB trading rule is the return of the buy-and-hold strategy for the HSI.

The out-of-sample test is also carried out for the TRB rules on three different time periods. The best time period, 50, 100 or 200 days, is chosen from one time period and

applies to next year's data. The average simulated returns for the 33 years will be collected and tested using the same procedure as in SMA strategy.

4. EMPIRICAL RESULTS

A. Sample Statistics

[Insert Table I here]

Table I contains summary statistics for the entire sample series and the four sub-periods for both mean daily returns and 10-day mean daily returns on the Hang Seng Index (HSI). The returns are calculated as raw returns of HSI². Panels A and B report the results of the mean daily returns (hereafter MDRs) and the 10-day mean returns (hereafter 10-day MDRs) respectively. The standard deviations (volatility) of the MDRs are substantially higher than the 10-day MDRs for all periods. The volatility of both sets of mean daily returns is quite stable across sub-periods except the first sub-periods ('72 – '78), which are substantially higher than the other three sub-periods and the full sample periods. The skewness of the MDRs for the full sample period, second and third sub-periods returns are significantly skewed to the left, and the first and four sub-periods returns significantly skewed to the right. Whereas the skewness of the 10-day MDRs for the full sample period, second, third and fourth sub-periods returns are significantly skewed to the left, and the first sub-period returns significantly skewed to the right. The skewness of the 10-day MDRs is, in general, slightly lower than the MDRs. The skewness in the third sub-period ('86 – '95) of both mean daily returns are exceptionally high and negative (-4.7817 for MDRs and -2.0291 for 10-day MDRs respectively), which causes the overall skewness to be negative in both sets of mean daily returns (skewnesses are -0.6301 and -0.4152 respectively). The high negative skewness is probably due to the sharp stock market drop in 5th June 1989 caused by the June 4th event. In general, the MDRs and 10-day MDRs are significantly skewed to the left.

Besides, the MDRs is highly leptokurtic (kurtosis is 19.8879) while the 10-day MDRs is just lightly leptokurtic (kurtosis is 3.9637). The high kurtosis in the MDRs is evidenced in all four sub-periods except the second sub-period ('79 – '85). To check

whether the kurtosis is caused by just a few outliers, sixteen outlier returns³ are deleted from the data set, and the skewness and kurtosis are recomputed for the reduced daily mean return series. The skewness and kurtosis for the reduced entire sample and the four sub-periods are similar to the full sample. In general, we can infer that the non-normality is not caused by the outliers. We also use log returns to compute the skewness and kurtosis and find that they are similar to those of the raw returns. Since we are doing out of sample test for trading rules, we need to use continuous daily returns. So, we still choose to use raw returns and include the outliers in the sample periods.

The serial correlations are generally very small except in the first sub-period ('72 – '78) of both of the mean daily returns. The two sets of returns exhibit very similar serial correlation patterns in the first five serial correlations. They both have significant positive first order serial correlation, $\rho(1)$. The higher order (second to five) serial correlations are mostly insignificant. The second and third order serial correlations are, in general, negative and insignificant while the fourth and fifth order serial correlations are insignificant and very small.

B. Empirical Results

In this section we report the empirical results for the traditional and out-of-sample test on the two trading strategies, simple moving average (SMA) and trading range break (TRB). We will report the finding of SMA in the first and second sub-sections (B.1. and B. 2.) and the results of TRB in the third sub-section (B. 3.). We are testing two different strategies under SMA, namely variable moving average strategy (VMA) and fix moving average strategy (FMA). We will discuss the VMA strategy first (B. 1.) and then the FMA strategy (B. 2.).

B.1. Variable Moving Average Strategy (VMA)

[Insert Table II here]

² The log differences of HSI are also calculated. Since the two results are very similar, we only report the results of the raw returns here.

³ The dates deleted are 27th Nov 1974 (Hong Kong dollars are allowed to float due to oil crisis), 12th and 13th August 1982 (property market crash), 26th October 1987 (Black Monday crash), 5th June 1989 (June 4th

Table II contains the result of the traditional test of Variable Moving Average (VMA) strategy over the full sample period and the four sub-periods. To save space, we only report the results for bandwidth zero percent. Column two presents the trading rules employed in the study. These rules are differing from each other by the length of the short and long moving average period and by the width of the band. For example (1, 50, 0) indicates that the short period is one day, the long period is fifty days, and the bandwidth is zero percent. Columns three and four report the number of buy and sell days generated by VMA. On average, for all twenty trading rules, there are 63.66% more buy days (4,963.30) than sell days (3,032.70). It is consistent with the upward-market trend of HSI during the sample period. Buy and sell average returns are reported separately in columns five and six. In Panel A, all buy returns are positive with only the (1, 50) trading rule returns statistically larger than the unconditional mean daily return (MDR at 0.0645% daily or 16.125% annually), at the 5% significant level. The average one-day buy return is 0.09% (about 22.50% per year), which is close to those reported by Bessembinder and Chan (1995) (0.084%). On the buy side, the trading rules (1, 50) are the best performing trading rules with the highest average returns and t-values among all reported trading rules.

On the sell side, all sell returns are mostly positive except the (1, 50) trading rules. The returns of the (1, 50) trading rule are statistically larger than the unconditional mean daily return at the 5% significant level. The average one-day sell return is 0%, which is slightly lower than those of Bessembinder and Chan (1995) (0.03%). On the sell side, again, the trading rules (1, 50) are the best performing trading rules with the highest average returns and t-values among all reported trading rules⁴.

When we perform tests with 1-day lag returns, we find that the average buy and sell return figures are slightly higher but very close to those of Bessembinder and Chan (1995). Our 1-day lag average buy and sell returns are 0.09% (22.50% annually) and 0.01% (2.50% annually) respectively while their returns are 0.079% (19.75% annually)

event), 23rd and 24th Oct 1997 (Asia financial crisis and property market collapse), 2nd February 1998 (property downturn) and 11th September 2001 (911 terrorist attack).

⁴ The negative returns under the (1, 50) trading rules turn positive when short selling strategy is used. Therefore, the (1, 50) trading rules are considered to be the best.

and -0.041% (-10.25% annually) respectively. The slight differences on both 0-day and 1-day lag returns may be due to difference test periods employed. We use a much longer test period (1972 – 2006, 35 years), sixteen years longer than those used by Bessembinder and Chan (1995) (1975 – 1991, 17 years).

Columns seven and eight present the probability of having a net gain in both buy and sell days. The probability of having a net gain ranges narrowly from 51% to 54% with most of the average buy and sell gain percentages come very close to each other. With these slightly above average probability rate, we can infer that the significant returns on buy and sell signals are not likely due to the inclusion of the sixteen extreme observations. The ninth and last column shows the difference between the mean daily buy and sell returns. The buy-sell differences are positive and significantly different from zero for the full sample period for only three strategies, (1, 50, 0), (1, 50, 1) and (1, 150, 0) at the 5% confidence level. For the four sub-periods (Table II Panel B), we only observe similar results in the first sub-period (1972 – 78). For the second sub-period (1979 – 85), only the buy-sell differences of the (1, 50) trading rules are positive and significantly difference from zero. For the last two sub-periods (1986 – 95 & 1996 – 06), no returns of trading rules are positive and significantly different from zero at the 5% confidence level. The full sample buy-sell difference ranges from 0.05% daily (12.50% annually) to 0.22% daily (55% annually). The average buy-sell difference for the twenty trading rules is positive (0.09% daily and 22.50% annually) and significantly different from zero at the 5% level of significance. This rejects the null hypothesis that the mean daily buy return is the same as the mean daily sell return. But it should be noted that the significant average buy-sell return is mainly caused by the significant buy-sell returns from two groups of trading rules, the (1, 50) and (1, 150) rules. Besides, we also observed that the introduction of bandwidth increases the spread between the buy and sell returns with the spread difference proportionally related to the bandwidth.

The results show that the profits are inconsistent across sub-periods. Most of the trading rules' buy and sell profits become insignificant in sub-periods. Even for the (1, 50) trading rules, their buy and sell profits remain significant only in the first two sub-periods (1972-78 and 1979-85). Besides, the magnitude of the returns decreases over the four

sub-periods. The trading rules with the highest returns in each sub-periods are always the (1, 50) rules. In general, we can infer from the results that the only trading rule, among the five we study, that can consistently outperform the market is rule (1, 50). Our results are different from the consistent sub-period results found in Brock et al. (1992). The results seem to suggest that the Hong Kong stock market becomes more efficient over time. One possible reason is that stock market integration leads to a more efficient dissemination of information over time.

[Insert Table III here]

Table III reports returns generated by the out-of-sample test. Since the results are very similar for different bandwidths, we only report the results for bandwidth zero and one. Full sample period results are shown in panel A. Same as the traditional test, all buy and short selling returns are statistically larger than the unconditional mean daily return. The out-of-sample tests generate 404 and 1,014 trading transactions for band zero and one percent respectively during the 35 years of study, which is averaging 12.24 and 30.72 trading transactions per year respectively. It is noteworthy to point out that the out-of-sample tests have consistently chosen only one single trading rule, namely (1, 50), for the full period of study.

The average daily buy return for the (1, 50, 0) trading rule is 0.15% (37.5% annually), and is substantially and significantly higher than the benchmark MDR of 0.063% (15.75% annually). For short selling, the average daily sell return of (1, 50, 0) is -0.07% (-17.5% annually), which is also significantly different from the benchmark MDR. The buy-sell daily return of (1, 50, 0) is 0.21% (52.50% annually), which is significantly greater than the daily benchmark MDR even at the 1% level. Panel B shows the results of each of the four non-overlapping sub-periods. Same as before we also report the results for bandwidth zero and one in the table. The sub-period tests service three purposes here. The first purpose is to test the consistency of the trading rules. The second purpose is to check which trading rule is the most useful and profitable rule in the twenty trading rules. The third and last purpose is to check whether the trading rule changes its usefulness with different testing periods and starting sub-period dates.

In Panel B, except the first sub-period, both buy and selling returns are insignificantly different from the benchmark unconditional mean daily return in the four sub-periods. The magnitude of the profit also decreases over time with profit in early sub-periods higher than those in the later sub-periods. For example, the buy and sell profits for (1, 50, 0) in the first sub-period ('72 – '78) are 0.27% and -0.23% respectively while those of the profits in the fourth sub-periods are 0.09% and -0.03% respectively. The magnitude of the profit in the fourth sub-period is very close to the benchmark daily MDR. Therefore, profit consistency for the buy and sell strategies is not held in the sub-periods. However, the buy-sell returns are significantly greater than the benchmark daily return in the first sub-period.

In sum, the out-of-sample test has helped us to identify a profitable trading strategy, the (1, 50) trading rule, which gives us returns statistically greater than the benchmark buy and hold trading strategy. However, the performance of the trading strategy does not seem to be consistent across sub-periods. The strategy works better in the early pre-1986 period when the stock market was supposed to be rather inefficient. But after 1986, when the four stock exchanges merged and the Hong Kong stock market started to grow, the profitable returns disappeared. The results seem to indicate that Hong Kong stock market was getting more and more efficient after 1986. Besides, we also try using different starting date for the sub-periods to perform the out-of-sample tests. The (1, 50) rule always keep emerging as the most useful and profitable trading rule no matter which starting dates (1972, 1979, 1986 or 1995) we use in the tests. In general, the (1, 50) trading rule is still the most profitable and useful trading rule among the twenty rules tested. It is profitable before 1986 but not after 1986. The latter finding contradicts some of the previous studies that returns are predictable in Hong Kong, which suggests that Hong Kong stock market may be efficient in its weak form after 1986.

B.2. Fix Moving Average Strategy

[Insert Table IV here]

Fix moving average trading strategy (FMA), a variation of VMA, further assumes that returns will be different for some time and it requires investor to stay in the same position after a crossing for a fix period of time, 10 days in our study. Table IV shows the findings of the full sample period and sub-period results. The results are similar with the findings of VMA (Table II). The different is that both buy and sell return figures are slightly lower than those of VMA (Table II). For the full sample period, only the buy and sell returns of the (1, 50) trading rules are significantly greater than the benchmark 10-day mean daily return. For example, the average buy and sell returns for (1, 50, 0) are 0.14% (35% annually) and -0.06% (-15%) respectively, which is significantly higher than the benchmark 10-day MDR of 0.063% (15.75% annually) at the 5% level. The buy and sell returns of other trading rules are all insignificantly different from the benchmark return. Similar to the VMA, the average buy and sell returns are insignificant different from zero (0.09% and 0.01% respectively) while the buy-sell returns are significantly different from zero with an average of 0.08% (20% annually) for the twenty trading rules. For the 1-day lag average buy and sell returns, the values remain insignificantly at 0.09% (22.50% annually) and 0.01% (2.50% annually), which are different slightly from the findings of Bessembinder and Chan (1995) (0.067% and -0.043% respectively).

The result of the FMA trading rules is expected. As all signals during the immediate nine trading days after the cross are ignored, the accuracy of FMA should be smaller than VMA. Even though the returns of FMA are slightly lower than those of the VMA, it seems that the information within the first nine trading days of the signal is still valuable to investors. By making use of this information, investors, utilizing the VMA strategy, can improve their annual trading profits by a few percentages. With the information within the nine trading days, investors' average buy-sell returns are on average higher in the VMA (0.09%) than those of the FMA (0.08%), which further indicates that the information within the first nine trading days is valuable to investors.

The results of sub-periods are shown in Table IV Panel B. In each of the four sub-periods, the pattern of t-values and return figures for both buy and sell more or less resembles than those of VMA except that the returns and t-values are slightly smaller. The average buy and sell returns of sub-periods decrease over time with the first sub-period having the highest average returns and last sub-period having the lowest returns.

Same as the VMA, only the (1, 50) trading rule generates significant profits in the first two sub-period, the pre-1986 sub-periods.

[Insert Table V here]

Out-of-sample test results are presented in Table V. With a zero percent trading band, the out-of-sample test requires investors to transact 232 times over the 35 years of study, which is approximately half of the VMA's. Over the full sample period, the out-of-sample test has again chosen only one single trading strategy, the (1, 50) rule, as the best trading rule across all twenty trading rules. The buy returns, except returns from trading strategy with 3% bandwidth, are significantly larger than the benchmark unconditional mean daily return at the 5% level. However, all sell returns are significantly different from the benchmark MDR at the 5% level. The mean return is the smallest when a 3% bandwidth is used but the difference between bands are small.

Same testing procedures are again repeated for each of the four sub-periods. Buy (sell) returns are positive (negative) and significantly larger than the benchmark unconditional mean daily return for the first sub-period ('72 – '78) except for the 3% bandwidth buy return. The second sub-period ('79 – '85) buy returns are all positively insignificantly difference from the benchmark return. For the sell returns, only the returns from the 0% trading strategy are significantly different from benchmark return. The rest of the sell returns are insignificant. The buy and sell returns in the next two sub-periods are all insignificantly different from the benchmark return.

From the FMA out-of-sample tests, we find that the trading rule (1, 50) is still the most profitable trading rule for all four sub-periods. But the buy returns are only significant in the first two sub-periods (pre-1986 sub-period), which is very similar to the result of the out-of-sample test in the VMA (Table III). From the results of the tests, both the FMA traditional and out-of-sample tests further confirms that trading rule (1, 50), on all four bandwidths, is the most profitable trading rule among the twenty rules we study in this paper. However, the (1, 50) trading rule works better before 1986. Using SMA trading rules after 1986, we cannot generate any significant profit over the benchmark

buy-and hold strategy, which indicates that the Hong Kong stock market is possibility more efficient after 1986.

B.3. Trading Range Break

[Insert Table VI here]

Full sample and sub-period results generated by Trading Range Break (TRB) rule are presented in Table VI. Over the 35 years of study, no mean buy and sell returns are statistically larger than the unconditional mean daily return at the 5% level. For the full sample (Panel A), the average daily buy and sell returns are 0.07% (17.5% annually) and 0.05% (12.5% annually) respectively. The buy-sell return is also the largest (0.10% daily or 25% annually) for the (50, 0) rule with very high t-value (2.25). The average buy-sell return for all twenty rules is 0.02% (5% annually). Both the (50, 0) and the average buy-sell returns are lower than the SMA (1, 50) rules' average return. The probability of having a gain in both buy and sell days are 0.53 and 0.51 respectively. All the buy, sell and buy-sell returns are insignificantly different from the benchmark return except the (50, 0) and (50, 1) buy-sell returns. It seems that the returns generated by TRB strategy are not that impressive and are lower than both of the VMA and FMA trading strategies.

For the four sub-periods (Panel B), we observe similar pattern with all buy, sell and buy-sell returns insignificantly different from the benchmark return. All the returns decrease over sub-periods with the early sub-periods having higher returns. This pattern is the same as those in the VMA and FMA tests. In general, the profitable buy and buy-sell returns are inconsistent across sub-periods and, unlike the FMA and VMA, all the profitable returns are insignificant.

[Insert Table VII here]

The results of the out-of-sample test are shown in Table VII. Same as SMA, the out-of-sample test method also consistently chooses only rule 50 days for all trade bandwidth during each of the 35 years of study. For zero trading bandwidth, the out-of-

sample test requires investors to make 97 transactions over the full sample period of study. All average buy and sell returns are statistically insignificantly different from the unconditional mean daily return at the 5% level. The average buy-sell returns are positive and significantly different from the unconditional MDR only for zero and one bandwidth strategies.

Panel B reports the result for each of the four sub-periods. The buy and sell returns are all insignificantly different from the unconditional mean daily return except the first sub-period while the buy-sell returns are insignificant for bandwidth three percent only. The out-of-sample results suggest that the TRB rule provides lower profitable returns than those of the VMA and FMA rules for the full sample and the four sub-periods. For example, the average buy and sell returns for the TRB full period sample (bandwidth zero and one) are both 0.10% while the corresponding buy-sell returns for the VMA rules are 0.15%, 0.14%, and for the FMS are 0.14% and 0.14% respectively.

From the traditional and out-of-sample tests of the VMA, FMA and TRB rules, we can observe the following results. First, the (1, 50) rule is the most profitable SMA trading rule that we can use in Hong Kong stock market among the twenty SMA rules we tested. The results further indicate that the VMA rule perform better than the FMA rule by including the information of the first nine days into investors' decision. Second, it suggests that the (1, 50) SMA rule is capable of capturing useful information for investors to predict the market movement of the Hong Kong stock market before 1986, but not after 1986. Whether this indicates that the Hong Kong stock market is weak form inefficient before 1986 and efficient after 1986, we cannot conclude from the tests in this paper. It needs further research to investigate the problem. Third, the (1, 50) SMA rules (both VMA and FMA rules) provide higher profits than the TRB rules. The two rules perform much better than the TRB rules, ranging up to 50%. Our results show that the (1, 50) SMA rule is the best rule among the trading rules we evaluated in this study. The only deficiency is that the rule is not consistent across sub-periods.

C. Transaction costs

In this section, we also consider the impact of transaction costs on the trading rules. The round-trip transaction costs of buying stocks range from 1.17% (29 Oct. 87 to

1 Apr. 91) to 0.74% (after 7 Apr. 2000) within our test period (source: Hong Kong Stock Exchange Fact Book 2004, Historical Section). Therefore, the average round-trip transaction cost is approximately 1%⁵ and we take this average transaction cost as the transaction cost in this study.

The impact of the transaction costs on trading profit is quite insignificant⁶. The net buy, sell and buy-sell net returns for full sample period of the three out of sample tests are reported in Table VIII. The net returns are, in general, still substantial for some trading rules even after taking into consideration of transaction costs. For example, the average buy, sell and buy-sell returns decrease by approximately 0.04%, 0.06% and 0.09% respectively for the OFS 0% VMA out of sample trading rule (Tables III and VIII). The buy returns remain significant for OFS 0% and 1% for VMA and FMA while the sell returns become insignificant. However, the buy-sell returns remain significant throughout the VMA and FMA out of sample tests. So, in general, the effect of transaction costs on VMA, FMA and TRB rules are insignificant.

5. CONCLUSIONS

In this paper, we investigate two simple but very popular trading rules, simple moving average and trading range break by looking at a series of data of Hang Seng Index over thirty-five years, from 1972 to 2006. We apply both bootstrap and out-of-sample tests to test the validity of the profitability of the trading rules. The out-of-sample test, which is not affected by the data-snooping biases problem, is used to check the usefulness of the traditional test.

In general, our results show that there is one trading rule, the (1, 50) rule, outperforms the market (HSI) over the 35 years of testing period and in the pre-1986 sub-periods of our study. The out-of-sample average buy and sell returns for zero bandwidth on VMA strategy are 0.15% and -0.07% (37.5% and -17.5% annually) respectively before transaction costs. The buy (sell) returns are significantly higher (lower) than the unconditional mean daily return of 15.75% per year. The 10-day FMA strategy products a lower average returns but still are way higher than the benchmark. The average

⁵ To make computation easier, we round up the average round-trip transaction cost from 0.955% to 1% and the one-way transaction cost to 0.5%.

bandwidth zero buy and sell FMA out-of-sample test returns are 0.14% and -0.06% (35% and -15% annually) respectively and are statistically different (higher or lower) than the benchmark return. These returns are also higher than the returns found in the US market (Brock et al., 1992) and Hong Kong market (Bessembinder and Chan, 1995). Besides, the VMA (1, 50) rule performs better than the FMA (1, 50) rule because it includes the information of the first nine days into investors' decision. With the flexibility of buy or sell within the first nine days, the VMA (1, 50) rule can generate 2.5% to 7.5% (annual) more profit than the FMA rule before transaction costs. However, the returns of the TRB rules are all small and insignificant.

The observation that trading strategies generate high abnormal average returns seems to suggest that the Hong Kong stock market is inefficient. This conclusion is a bit premature and, therefore, we further explore four alternative explanations for the observation. First, the positive abnormal return may possibly be caused by the different risk in different trading strategies. After applying out-of-sample test to the returns, we find that it is unlikely that the difference risk in different trading strategies can explain the abnormal returns.

The second explanation for the high positive abnormal returns is the positive autocorrelation in the Hong Kong stock market. But the small positive autocorrelation we found (0.0826) does not seem to be able to explain the high buy and sell average returns. Part of the reasons that the autocorrelation is small is due to the nature of the HSI. Since the constituent stocks in the HSI are all actively traded stocks, there is no "thin trading" or "nonsynchronous" problem in the index. This nature of the HSI would reduce the autocorrelation in the index. Besides, the high average buy-sell returns (e.g., 0.15% daily or 37.5% annual for band zero VMA out-of-sample test) are also difficult to explain by risk. We do not see any substantial change in volatility between buy and sell signals. The standard deviations within the period on the both signals remain pretty much the same. So, it does not seem likely that the large difference can be explained by increase in risk due to change in signals.

⁶ The per day transaction cost is calculated by multiplying the one-way (for buy or sell) transaction cost with the number of trades $[N(\text{Trades})]$ and divided it by the number of buy $[N(\text{Buy})]$ or sell $[N(\text{Sell})]$.

Even though the high returns in our test are unlikely due to autocorrelation in stock prices, there are still two other possible explanations. One possible explanation is the transaction costs and the other is time-varying risk premium. In our case, the average returns after transaction costs are still high. The trading costs in Hong Kong, which is approximately one percent per round trip, will not affect the average returns too much. In fact, the transaction costs only reduce the average returns by 0.04% to 0.06% (daily) on VMA and FMA rules. So, the transaction costs can only explain a small fraction of the high average returns.

Finally, we find that the trading rules work better before 1986 which suggests that the Hong Kong stock market becomes more efficient over time. One possible reason is due to the stock market integration which leads to a more efficient dissemination of information over time.

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Table I
Descriptive Statistic

Results are shown for the full period and 4 non-overlapping sub-periods. Returns are measured as percentage differences of the Hang Seng Index. 10-day return is the average of 10 non-overlapping daily return. N is the number of observations. $\rho(i)$ is the estimated i days lag autocorrelation. Figures marked in bold letter are at least 10% significant.

Panel A: Daily return					
Raw Return	Full Sample ('72 - '06)	1st sub- period ('72 - '78)	2nd sub- period ('79 - '85)	3rd sub- period ('86 - '95)	4th sub- period ('96 - '06)
N	8143	1728	1721	2227	2467
Mean	0.0645%	0.0503%	0.0914%	0.0845%	0.0376%
Stdev	1.9201%	2.4063%	1.8900%	1.6872%	1.7478%
Skewness	-0.630091	0.532001	-0.326381	-4.781662	0.450633
Kurtosis	19.887910	5.648920	2.877823	84.738283	11.390759
Serial Correlations					
$\rho(1)$	0.0826*	0.1609*	0.0473*	0.0650*	0.0274
$\rho(2)$	-0.0167	-0.0099	0.0132	-0.0259	-0.0368
$\rho(3)$	-0.0040	-0.0123	0.0010	-0.0041	-0.0021
$\rho(4)$	0.0067	0.0099	0.0031	0.0068	0.0038
$\rho(5)$	-0.0015	0.0072	0.0015	-0.0034	-0.0046
Panel B: 10-Day return					
Raw Return	Full Sample ('72 - '06)	1st sub- period ('72 - '78)	2nd sub- period ('79 - '85)	3rd sub- period ('86 - '95)	4th sub- period ('96 - '06)
N	809	167	172	223	247
Mean	0.000657	0.000520	0.000890	0.000855	0.000408
Stdev	0.006792	0.009384	0.007029	0.005709	0.005308
Skew	-0.415163	0.147686	-0.475270	-2.029093	-0.353655
Kurtosis	3.963706	2.072920	0.621576	10.983236	1.770637
Serial Correlations					
$\rho(1)$	0.1260*	0.2477*	0.0548*	0.0701*	0.0611*
$\rho(2)$	-0.0138	0.1371*	0.0805*	-0.0700	-0.0034
$\rho(3)$	-0.0078	0.0368	0.0078	-0.0114	-0.0009
$\rho(4)$	-0.0018	0.0253	0.0098	0.0017	0.0014
$\rho(5)$	0.0025	0.0256	0.0096	-0.0010	0.0009

* significant at 1% level;

** significant at 5% level;

*** significant at 10% level

Table II
Variable Moving Average Trading Strategy

The test period is from 1972 to 2006. Rules are classified as (short, long, band). Short represents the length of the short moving average and long represents the length of the long moving averages. Band is the percentage difference between the long and short moving averages that requires to generate a signal. "N(buy)" and "N(sell)" represent the total number of buy and sell days generated within the period. Numbers which sits underneath the percentage return is the corresponding t-ratios. The t-value is testing the difference of the mean buy and sell returns from the unconditional mean return, and the buy-sell from zero. "Buy>0" and "Sell>0" measures the probability of getting a positive return in a buy and sell day.

Full Sample

Panel A

Period	Test	N(Buy)	N(Sell)	Buy	Sell	Buy>0	Sell>0	Buy-sell
1972-2006	(1,50,0)	4872	3124	0.15% 2.34**	-0.07% -3.33*	0.54	0.52	0.22% 4.92*
	(1,150,0)	5079	3059	0.10% 1.17	0.00% -1.61	0.53	0.51	0.11% 2.41**
	(1,200,0)	4856	2967	0.08% 0.45	0.01% -1.32	0.53	0.51	0.07% 1.62
	Average			0.09% 0.85	0.00% -1.58			0.09% 2.12**

Four sub-periods

Panel B

Period	Test	N(Buy)	N(Sell)	Buy	Sell	Buy>0	Sell>0	Buy-sell
1972-78	(1,50,0)	799	771	0.27% 2.04**	-0.21% -2.39**	0.53	0.56	0.48% 3.84*
	(1,150,0)	963	749	0.16% 1.06	-0.08% -1.21	0.52	0.54	0.24% 2.01**
	(1,200,0)	723	674	0.04% -0.08	-0.07% -1.09	0.50	0.55	0.11% 0.95
1979-85	(1,50,0)	1098	620	0.17% 1.11	-0.06% -1.74***	0.55	0.52	0.23% 2.47**
	(1,150,0)	1080	562	0.15% 0.76	-0.02% -1.25	0.54	0.50	0.17% 1.74***
	(1,200,0)	1045	660	0.12% 0.40	0.03% -0.72	0.54	0.49	0.09% 0.98
1986-95	(1,50,0)	1539	677	0.12% 0.70	-0.01% -1.21	0.56	0.50	0.13% 1.65***
	(1,150,0)	1611	579	0.07% -0.19	0.12% 0.51	0.54	0.47	-0.05% -0.62
	(1,200,0)	1615	573	0.08% -0.12	0.11% 0.37	0.54	0.47	-0.04% -0.44
1996-2006	(1,50,0)	1422	1026	0.08% 0.76	-0.02% -0.88	0.51	0.51	0.10% 1.42
	(1,150,0)	1385	1028	0.08% 0.66	-0.02% -0.88	0.52	0.52	0.10% 1.34
	(1,200,0)	1431	982	0.08% 0.67	-0.02% -0.94	0.52	0.52	0.10% 1.40

* significant at 1% level;

** significant at 5% level;

*** significant at 10% level

Table III
Variable Moving Average - Out of Sample Test result

The test period is from 1972 to 2006. Rules are classified as (short, long). Short represents the length of the short moving average and long represents the length of the long moving averages. OFS 0% represents Out of Sample Test with a zero percentage band. Band is the percentage difference between the long and short moving averages the requires to generate a signal. "N(trade)" represents the number of trade or transaction made. "N(buy)" and "N(sell)" represent the total number of buy and sell days generated within the period. N(Rules) is the total number of trading rules chosen by Out of Sample Test over the period of study. Numbers which sits underneath the percentage return is the corresponding t-value (in parenthesis) is testing the difference of the mean buy and sell returns from the unconditional mean return, and the buy-sell from zero.

Panel A: Full period results (1972 - 2006)

Test	N(Trades)	N(Buy)	N(Sell)	N(Rules)	Most chosen rule	Buy	Sell	Buy-Sell
OFS 0%	404	4866	3122	1	(1,50)	0.15% (2.47**)	-0.07% (-3.31*)	0.21% (5.01*)
OFS 1%	206	4931	3057	1	(1,50)	0.14% (2.32**)	-0.06% (-3.18*)	0.21% (4.76*)
OFS 2%	156	4954	3031	1	(1,50)	0.13% (1.98**)	-0.04% (-2.72*)	0.18% (4.07*)
OFS 3%	134	4950	3035	1	(1,50)	0.12% (1.66)	-0.03% (-2.26**)	0.15% (3.40*)
Average						0.10% 1.07	0.00% -1.67	0.10% 2.37**

Panel B: Sub-period results

(a) Subperiod 1: 1972 - 1978

Test	N(Trades)	N(Buy)	N(Sell)	N(Rules)	Most chosen rule	Buy	Sell	Buy-Sell
OFS 0%	65	799	770	1	(1,50)	0.27% (2.06**)	-0.23% (-2.57*)	0.50% (4.02*)
OFS 1%	39	802	767	1	(1,50)	0.25% (1.83***)	-0.21% (-2.53**)	0.45% (3.62*)
OFS 2%	31	824	742	1	(1,50)	0.23% (1.69***)	-0.20% (-2.25**)	0.43% (3.42*)
OFS 3%	27	804	762	1	(1,50)	0.22% (1.59)	-0.18% (-2.08**)	0.40% (3.19*)

(b) Sub-period 2: 1979 -1985

OFS 0%	87	1098	618	1	(1,50)	0.17% (1.12)	-0.05% (-1.63)	0.22% (2.38*)
OFS 1%	37	1086	630	1	(1,50)	0.19% (1.36)	-0.08% (-1.94***)	0.27% (2.85*)
OFS 2%	35	1062	652	1	(1,50)	0.16% (0.93)	-0.02% (-1.27)	0.18% (1.91***)
OFS 3%	31	1076	638	1	(1,50)	0.14% (0.67)	0.01% (-0.93)	0.13% (1.38)

(c) Sub-period 3: 1986 - 1995

OFS 0%	103	1537	678	1	(1,50)	0.13% (0.83)	0.04% (-0.69)	0.09% (1.26)
OFS 1%	59	1566	643	1	(1,50)	0.11% (0.57)	0.07% (-0.22)	0.04% (0.61)
OFS 2%	43	1590	618	1	(1,50)	0.10% (0.40)	0.09% (0.10)	0.01% (0.18)
OFS 3%	33	1609	581	1	(1,50)	0.10% (0.36)	0.10% (0.22)	0.00% (0.03)

(d) Sub-period 4: 1996 -2006

OFS 0%	149	1418	1026	1	(1,50)	0.09% (0.84)	-0.03% (-1.02)	0.11% (1.61)
OFS 1%	71	1457	985	1	(1,50)	0.09% (0.86)	-0.03% (-1.09)	0.12% (1.69***)
OFS 2%	47	1457	981	1	(1,50)	0.09% (0.90)	-0.03% (-1.13)	0.12% (1.75***)
OFS 3%	43	1422	1016	1	(1,50)	0.07% (0.65)	-0.01% (-0.77)	0.08% (1.22)

* significant at 1% level;

** significant at 5% level;

*** significant at 10% level

Table IV**Fix Moving Average Trading Strategy**

The test period is from 1972 to 2006. Rules are classified as (short, long, band). Short represents the length of the short moving average and long represents the length of the long moving averages. Band is the percentage difference between the long and short moving averages that requires to generate a signal. "N(buy)" and "N(sell)" represent the total number of buy and sell days generated within the period. Numbers which sits underneath the percentage return is the corresponding t-ratios. The t-value is testing the difference of the mean buy and sell returns from the 10-day unconditional mean return, and the buy-sell from zero. "Buy>0" and "Sell>0" measures the probability of getting a positive return in a buy and sell day.

Full Sample**Panel A**

Period	Test	N(Buy)	N(Sell)	Buy	Sell	Buy>0	Sell>0	Buy-sell
1972-2006	(1,50,0)	4742	3254	0.14% 2.19**	-0.06% -2.99*	0.54	0.51	0.20% 4.50*
	(1,150,0)	5015	3123	0.11% 1.36	-0.01% -1.84***	0.53	0.51	0.12% 2.78*
	(1,200,0)	4836	2987	0.07% 0.16	0.03% -0.92	0.53	0.50	0.04% 1.00
	Average			0.09% 0.74	0.01% -1.48			0.08% 1.94***

Four sub-periods**Panel B**

Period	Test	N(Buy)	N(Sell)	Buy	Sell	Buy>0	Sell>0	Buy-Sell
1972-78	(1,50,0)	795	775	0.27% 2.07**	-0.21% -2.40**	0.53	0.56	0.49% 3.87*
	(1,150,0)	953	759	0.18% 1.32	-0.11% -1.48	0.53	0.55	0.29% 2.47**
	(1,200,0)	732	665	0.02% -0.27	-0.05% -0.90	0.50	0.54	0.07% 0.61
1979-85	(1,50,0)	1095	623	0.19% 1.36	-0.09% -2.10**	0.55	0.52	0.29% 3.01*
	(1,150,0)	1063	579	0.14% 0.68	-0.01% -1.09	0.54	0.50	0.15% 1.52
	(1,200,0)	1047	658	0.11% 0.24	0.05% -0.51	0.53	0.48	0.06% 0.66
1986-95	(1,50,0)	1473	743	0.11% 0.50	0.03% -0.79	0.55	0.49	0.09% 1.12
	(1,150,0)	1591	599	0.08% -0.02	0.10% 0.17	0.55	0.48	-0.01% -0.18
	(1,200,0)	1587	601	0.07% -0.22	0.13% 0.55	0.54	0.47	-0.05% -0.68
1996-2006	(1,50,0)	1365	1083	0.06% 0.33	0.02% -0.33	0.50	0.50	0.04% 0.57
	(1,150,0)	1368	1045	0.08% 0.69	-0.02% -0.89	0.52	0.52	0.10% 1.37
				-0.04	-0.02			-0.02
	(1,200,0)	1428	985	0.07% 0.52	-0.01% -0.74	0.52	0.52	0.08% 1.10

* significant at 1% level;

** significant at 5% level;

*** significant at 10% level

Table V**Fix Moving Average - Out of Sample Test result**

The test period is from 1972 to 2006. Rules are classified as (short, long). Short represents the length of the short moving average and long represents the length of the long moving averages. OFS 0% represents Out of Sample Test with a zero percentage band. Band is the percentage difference between the long and short moving averages that requires to generate a signal. "N(trade)" represents the number of trade or transaction made. "N(buy)" and "N(sell)" represent the total number of buy and sell days generated within the period. N(Rules) is the total number of trading rules chosen by Out of Sample Test over the period of study. Numbers which sits underneath the percentage return is the corresponding t-ratios. The t-value (in parenthesis) is testing the difference of the mean buy and sell returns from the 10-day unconditional mean return, and the buy-sell from zero.

Panel A: Full period results (1972 – 2006)

Test	N(Trades)	N(Buy)	N(Sell)	N(Rules)	Most chosen rule	Buy	Sell	Buy-Sell
OFS 0%	232	4741	3247	1	(1,50)	0.14% (2.32**)	-0.05% (-2.98*)	0.20% (4.59*)
OFS 1%	182	4844	3144	1	(1,50)	0.14% (2.15**)	-0.05% (-2.86*)	0.19% (4.34*)
OFS 2%	146	4895	3090	1	(1,50)	0.13% (2.02**)	-0.04% (-2.72*)	0.18% (4.10*)
OFS 3%	130	4922	3063	1	(1,50)	0.12% (1.63)	-0.02% (-2.20**)	0.14% (3.31*)
Average						0.10% 0.98	0.00% -1.58	0.10% 2.22**

Panel B: Sub-period results**(A) Subperiod 1: 1972 – 1978**

Test	N(Trades)	N(Buy)	N(Sell)	N(Rules)	Most chosen rule	Buy	Sell	Buy-Sell
OFS 0%	41	795	774	1	(1,50)	0.27% (2.09**)	-0.23% (-2.58*)	0.50% (4.05*)
OFS 1%	33	802	767	1	(1,50)	0.26% (1.95***)	-0.22% (-2.47**)	0.48% (3.83*)
OFS 2%	29	821	745	1	(1,50)	0.24% (1.76***)	-0.20% (-2.32**)	0.44% (3.55*)
OFS 3%	27	805	761	1	(1,50)	0.21% (1.45)	-0.16% (-1.94***)	0.37% (2.94*)

(B) Sub-period 2: 1979 – 1985

OFS 0%	39	1095	621	1	(1,50)	0.19% (1.37)	-0.08% (-2.00**)	0.28% (2.92*)
OFS 1%	35	1088	628	1	(1,50)	0.18% (1.23)	-0.06% (-1.76***)	0.24% (2.59*)
OFS 2%	35	1059	655	1	(1,50)	0.15% (0.78)	0.00% (-1.06)	0.15% (1.59)
OFS 3%	31	1076	638	1	(1,50)	0.14% (0.67)	0.01% (-0.93)	0.13% (1.38)

(C) Sub-period 3: 1986 – 1995

OFS 0%	69	1470	745	1	(1,50)	0.12% (0.60)	0.07% (-0.26)	0.05% (0.70)
OFS 1%	57	1506	703	1	(1,50)	0.10% (0.25)	0.11% (0.35)	-0.01% (-0.15)
OFS 2%	39	1559	649	1	(1,50)	0.11% (0.48)	0.08% (-0.04)	0.03% (0.38)
OFS 3%	31	1590	600	1	(1,50)	0.11% (0.44)	0.09% (0.07)	0.02% (0.23)

(D) Sub-period 4: 1996 – 2006

OFS 0%	83	1367	1077	1	(1,50)	0.06% (0.42)	0.01% (-0.46)	0.05% (0.76)
OFS 1%	57	1428	1014	1	(1,50)	0.08% (0.79)	-0.02% (-0.97)	0.11% (1.53)
OFS 2%	43	1435	1003	1	(1,50)	0.09% (0.95)	-0.04% (-1.17)	0.13% (1.83)
OFS 3%	41	1412	1026	1	(1,50)	0.07% (0.65)	-0.01% (-0.76)	0.08% (1.22)

* significant at 1% level;

** significant at 5% level;

*** significant at 10% level

Table VI
Trading Range Break Trading Strategy

The test period is from 1972 to 2006. Rules are classified as (short, long, band). Short represents the length of the short moving average and long represents the length of the long moving averages. Band is the percentage difference between the long and short moving averages that requires to generate a signal. "N(buy)" and "N(sell)" represent the total number of buy and sell days generated within the period. Numbers which sits underneath the percentage return is the corresponding t-ratios. The t-value is testing the difference of the mean buy and sell returns from the unconditional mean return, and the buy-sell from zero. "Buy>0" and "Sell>0" measures the probability of getting a positive return in a buy and sell day.

Full Sample

Panel A

Period	Test	N(Buy)	N(Sell)	Buy	Sell	Buy>0	Sell>0	Buy-Sell
1972-2006	(50,0)	4983	3136	0.10%	0.00%	0.53	0.51	0.10%
				1.10	-1.50			2.25**
	(150,0)	4809	2979	0.05%	0.06%	0.53	0.50	-0.01%
				-0.28	-0.11			-0.12
	(200,0)	5089	2620	0.07%	0.04%	0.53	0.51	0.03%
Average				0.17	-0.56			0.67
				0.07%	0.05%			0.02%
				0.07	-0.33			0.33

Four sub-periods

Panel B

Period	Test	N(Buy)	N(Sell)	Buy	Sell	Buy>0	Sell>0	Buy-Sell
1972-78	(50,0)	946	747	0.19%	-0.13%	0.53	0.56	0.32%
				1.40	-1.63			2.62*
	(150,0)	746	616	0.05%	-0.06%	0.50	0.55	0.11%
				0.00	-0.88			0.81
	(200,0)	814	469	0.04%	-0.02%	0.49	0.52	0.06%
				-0.06	-0.54			0.50
1979-85	(50,0)	974	733	0.10%	0.07%	0.53	0.48	0.03%
				0.09	-0.24			0.29
	(150,0)	418	637	-0.03%	0.09%	0.50	0.48	-0.12%
				-1.10	0.01			-1.01
	(200,0)	527	527	0.02%	0.08%	0.52	0.50	-0.06%
				-0.71	-0.07			-0.55
1986-95	(50,0)	1610	581	0.09%	0.06%	0.54	0.47	0.03%
				0.19	-0.26			0.38
	(150,0)	1202	583	0.00%	0.21%	0.52	0.45	-0.21%
				-1.36	1.53			-2.44
	(200,0)	1267	518	0.04%	0.12%	0.53	0.48	-0.08%
				-0.65	0.46			-0.90
Period	Test	N(Buy)	N(Sell)	Buy	Sell	Buy>0	Sell>0	Buy-Sell
1996-2006	(50,0)	4983	3136	0.10%	0.00%	0.53	0.51	0.10%
				0.37	-0.44			0.70
	(150,0)	4809	2979	0.05%	0.06%	0.53	0.50	-0.01%
				0.13	-0.31			0.38
	(200,0)	5089	2620	0.07%	0.04%	0.53	0.51	0.03%
				0.42	-0.73			1.01

* significant at 1% level;

** significant at 5% level;

*** significant at 10% level

Table VII**Trading Range Break - Out of Sample Test result**

The test period is from 1972 to 2006. Rules are classified as (short, long). Short represents the length of the short moving average and long represents the length of the long moving averages. OFS 0% represents Out of Sample Test with a zero percentage band. Band is the percentage difference between the long and short moving averages the requires to generate a signal. "N(trade)" represents the number of trade or transaction made. "N(buy)" and "N(sell)" represent the total number of buy and sell days generated within the period. N(Rules) is the total number of trading rules chosen by Out of Sample Test over the period of study. Numbers which sits underneath the percentage return is the corresponding t-value (in parenthesis) is testing the difference of the mean buy and sell returns from the unconditional mean return, and the buy-sell from zero.

Panel A: Full period results (1972 - 2006)

Test	N(Trades)	N(Buy)	N(Sell)	N(Rules)	Most chosen rule	Buy	Sell	Buy-Sell
OFS 0%	97	4974	3137	1	(1,50)	0.10% (1.11)	0.01% (-1.31)	0.09 (2.09**)
OFS 1%	77	4877	3212	1	(1,50)	0.10% (1.07)	0.02% (-1.21)	0.08% (1.96**)
OFS 2%	54	4058	3984	1	(1,50)	0.09% (0.61)	0.04% (0.55)	0.04% (1.00)
OFS 3%	26	1883	6057	1	(1,50)	0.09% (0.62)	0.05% (-0.40)	0.04% (0.82)
Average						0.08%	0.05%	0.03%
						0.29	-0.40	0.56

Panel B: Sub period results**(A) Sub-period 1: 1972 - 1978**

Test	N(Trades)	N(Buy)	N(Sell)	N(Rules)	Most chosen rule	Buy	Sell	Buy-Sell
OFS 0%	18	946	746	1	(1,50)	0.19% (1.41)	-0.15% (-1.81***)	0.34% (2.80*)
OFS 1%	14	882	788	1	(1,50)	0.20% (1.41)	-0.13% (-1.67***)	0.32% (2.67*)
OFS 2%	10	602	1021	1	(1,50)	0.21% (1.35)	-0.07% (-1.25)	0.28% (2.23*)
OFS 3%	8	403	1118	1	(1,50)	0.15% (0.72)	-0.03% (-0.83)	0.18% (1.25)

(B) Sub-period 2: 1979 - 1985

OFS 0%	23	974	731	1	(1,50)	0.10% (0.09)	0.08% (-0.13)	0.02% (0.19)
OFS 1%	19	992	711	1	(1,50)	0.11% (0.18)	0.07% (-0.29)	0.04% (0.41)
OFS 2%	15	1067	636	1	(1,50)	0.08% (-0.19)	0.11% (0.21)	-0.03% (-0.34)
OFS 3%	10	546	1042	1	(1,50)	-0.03% (-1.27)	0.16% (0.86)	-0.18% (-1.84)

(C) Sub-period 3: 1986 - 1995

OFS 0%	27	1602	588	1	(1,50)	0.09% (0.19)	0.12% (0.54)	-0.03% (-0.39)
OFS 1%	19	1611	576	1	(1,50)	0.10% (0.41)	0.10% (0.17)	0.01% (0.11)
OFS 2%	13	1019	1163	1	(1,50)	0.10% (0.21)	0.11% (0.48)	-0.01% (-0.22)
OFS 3%	6	557	1481	1	(1,50)	0.12% (0.55)	0.09% (0.03)	0.04% (0.50)

(D) Sub-period 4: 1996 - 2006

OFS 0%	29	1413	1005	1	(1,50)	0.06% (0.36)	0.01% (-0.44)	0.05% (0.69)
OFS 1%	25	1350	1066	1	(1,50)	0.04% (-0.02)	0.04% (-0.02)	0.00% (0.01)
OFS 2%	16	1323	1093	1	(1,50)	0.04% (0.00)	0.03% (-0.04)	0.00% (0.04)
OFS 3%	2	377	1162	1	(1,50)	0.16% (1.31)	0.02% (-0.27)	0.14% (1.45)

* significant at 1% level;

** significant at 5% level;

*** significant at 10% level

Table VIII**Net return of Out of Sample Test**

The test period is from 1972 to 2006. Rules are classified as (short, long). Short represents the length of the short moving average and long represents the length of the long moving averages. OFS 0% represents Out of Sample Test with a zero percentage band. Band is the percentage difference between the long and short moving averages that requires to generate a signal. "N(trade)" represents the number of trade or transaction made. "N(buy)" and "N(sell)" represent the total number of buy and sell days generated within the period. N(Rules) is the total number of trading rules chosen by Out of Sample Test over the period of study. Numbers which sits underneath the percentage return is the corresponding t-value is testing the difference of the mean buy and sell returns from the unconditional mean return, and the buy-sell from zero. All profits are net of transaction costs (one-way cost = 0.5% and round trip cost = 1%).

VMA: Full period result (1972 - 2006)

Test	N(Trades)	N(Buy)	N(Sell)	N(Rules)	Most chosen rule	Buy	Sell	Buy-Sell
OFS 0%	404	4866	3122	1	(1,50)	0.11% 1.81***	-0.01% -0.47	0.12% 2.86*
OFS 1%	206	4931	3057	1	(1,50)	0.12% 1.99**	-0.03% -1.59	0.15% 3.40*
OFS 2%	156	4954	3031	1	(1,50)	0.11% 1.74***	-0.02% -1.36	0.13% 2.94*
OFS 3%	134	4950	3035	1	(1,50)	0.11% 1.52	-0.01% -1.51	0.12% 2.72*

FMA: Full period result 1972 - 2006

Test	N(Trades)	N(Buy)	N(Sell)	N(Rules)	Most chosen rule	Buy	Sell	Buy-Sell
OFS 0%	232	4741	3247	1	(1,50)	0.12% 1.99**	-0.01% -0.60	0.13% 2.98*
OFS 1%	182	4844	3144	1	(1,50)	0.12% 1.84***	-0.02% -1.14	0.14% 3.20*
OFS 2%	146	4895	3090	1	(1,50)	0.12% 1.86***	-0.02% -1.36	0.14% 3.19*
OFS 3%	130	4922	3063	1	(1,50)	0.11% 1.49	0.00% -0.00	0.11% 2.60*

TRB: Full period result 1972 - 2006

Test	N(Trades)	N(Buy)	N(Sell)	N(Rules)	Most chosen rule	Buy	Sell	Buy-Sell
OFS 0%	97	4974	3137	1	(1,50)	0.09% 0.99	0.03% 1.31	0.06% 1.39
OFS 1%	77	4877	3212	1	(1,50)	0.09% 0.96	0.03% 1.82	0.06% 1.48
OFS 2%	54	4058	3984	1	(1,50)	0.08% 0.54	0.05% 0.66	0.03% 0.75
OFS 3%	26	1883	6057	1	(1,50)	0.08% 0.55	0.05% -0.38	0.03% 0.62

* significant at 1% level;

** significant at 5% level;

*** significant at 10% level